$I = \frac{dQ}{dt}$, $dI = \bar{j}\,\overline{dS}$, $dQ = \rho dV = \rho\, dS\, dl$

$dI = \frac{dQ}{dt} = \rho v dS$; $j = \rho v$; $\bar{j} = qn\bar{v}$

$\rho = \frac{Q}{V} = \frac{Nq}{V} = nq$; $I = \oint_S \bar{j}\,\overline{dS}$ локально 

$-\frac{dq}{dt} = \oiint_S \bar{j} \cdot \overline{dS}$, $\rightarrow$ $-\frac{d\rho}{dt} = \bar{\nabla} \cdot \bar{j}$

$\left(\lim_{\Delta V \to 0} \frac{dq}{dt\,\Delta V} = \lim_{\Delta V \to 0} \frac{\oint \bar{j}\,\overline{dS}}{\Delta V} \right)$

$\bar{\nabla} \cdot \bar{j} = 0$ – стационарность

$U = IR$; $U = El$; $I = jS$; $R = \rho \frac{l}{S}$ $\Rightarrow$ $E = \rho j = \frac{j}{\sigma}$